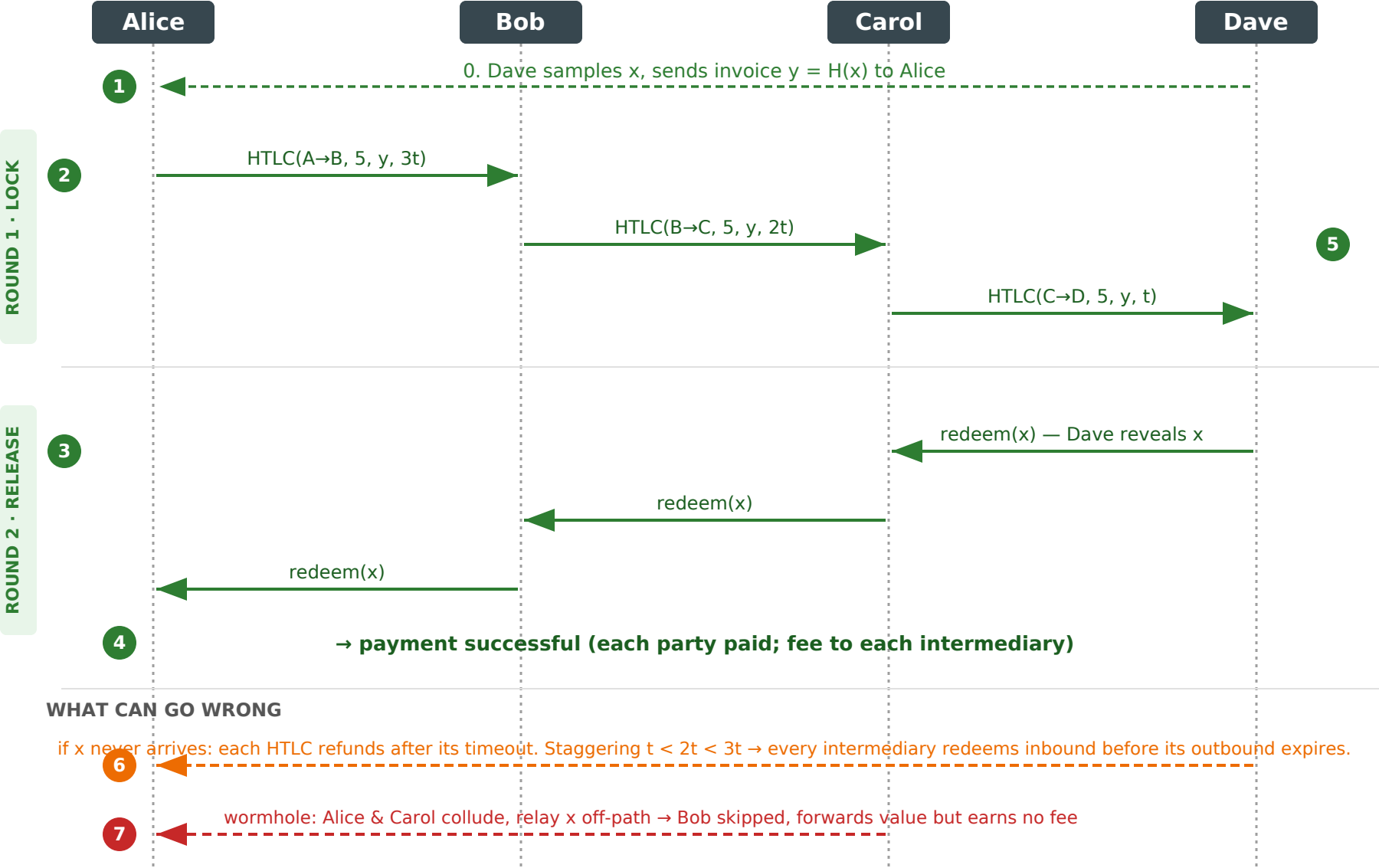


Multi-hop payment: Alice → Bob → Carol → Dave

annotated with multihop.spthy lemmas — see rail →



8 STAGGERED TIMELOCKS (block height →)



Rough draft — multihop.spthy lemmas mapped to the 4-party flow. Next: cltv_blocks / Clock / timeout.

LEMMA · multihop.spthy

- 1 Invoice & preimage authentication
Invoice_Authenticates_Settlement · Preimage_Secret_Until_Release · Forged_Invoice_Requires_Key_Compromise · Invoice_Released ✓
- 2 Lock causality (forward needs previous)
Forward1_Requires_Offer · Forward2_Requires_Forward1 · HTLC_On_Opened_Channel ✓
- 3 Release causality (redeem needs preimage)
Fulfill_Requires_Forward2 · Claim_Requires_Release · Settle_Requires_Receiver_Release ✓
- 4 Atomicity + non-vacuity
Payment_Atomicity_Under_Liveness ✓
Multihop_Payment_Possible ✓ (witness)
- 5 Value & fee conservation
EndToEnd_Value_Conservation · Fee_Conservation_Hop1/2 · Fee_Strictly_Positive_Hop1/2 · Receiver_Paid_Invoice_Amount ✓
- 6 Refund / griefing
Refund_Requires_Timeout · Loss_Requires_Inaction ✓
Honest_Intermediary_Refunds ✓ (reachable refund/DoS)
- 7 Wormhole fee theft
Wormhole_Fee_Theft_Reachable ✓
Wormhole_Steals_Exactly_The_Fees ✓ (theft = path fees)
- 8 Intermediary safety (under liveness T2b)
Intermediary_Never_Loses_Under_Liveness ✓
Timeout_race_Blocked ✓ (race blocked with T2b)

✓ = machine-checked in multihop.spthy (43/43).
green = all-traces safety · orange = benign/reachable
red = attack shown reachable (by design, no PTLC).
Channel-layer lemmas (open/update/close/punish) live on the Alice↔Bob figure.